Service Cutter: A Systematic Approach to Service Decomposition

Source: M. Aiello *et al.* "Service Cutter: A Systematic Approach to Service Decomposition," *Proceedings of the 5th European Conference on Service-Oriented and Cloud Computing (ESOCC 2016)*, 2016. [Online]. Available: http://link.springer.com/10.1007/978-3-319-44482-6_12

Introduction (What is the Paper about?)

- defining **Coupling Strategies in a catalog**
- present a tool called **Service Cutter** using that coupling strategies
- present useful definitions for the field

Methods

- **Service Cutter** extracts information from software engineering artifacts such as Domain Models and Use Cases
  - represent them as an uni-directed, weighted graph
  - find and score densely connected clusters
- **Validation Activities** including *Prototyping, Action Research* (?) and *Two Case Studies*
  - Performance Measurements
  - Feedback from members of the target audience in industry and academia

Scientific Benefit

- Collect **Coupling Strategies in a catalog** distilled from literature, industry and workshops
  - Catalog can serve to establish common terminology in discussions and documentation
- Introduced **Service Cutter** a knowledge management method and supporting tool framework assisting software architects when they making service design decisions
  - Can also suggest service decomposition candidates
- Most, but not all test scenarios resulted in appropriate service cuts
- Members from the target audience in industry and academia acknowledge that presented coupling criteria catalog and tool-supported service decomposition method have the potential to assist service architect's design decisions

Challenges / Future Work

Limitations

- not intended to fully automate Service Decomposition but rather support it
- collection might not yet be complete
  - (Own) Requirements on software might change in the future or might be weighted differently

Paper Outline

1. [X] Introduction
2. [X] Scopes the context and defines terminology
3. [X] Presentation of the Coupling Criteria Catalog
4. [ ] Definition of a novel service decomposition process and tool architecture
5. [ ] Presentation of an Implementation *(Service Cutter)* and validation
   - includes two case studies and performance measurements
6. [ ] Discussion of Strengths and Weaknesses of Service Cutter
7. [ ] Conclusion and Future Work

Own Comments

- (+) Key paper
- (+) Paper serves as foundation of the field of *Service Decomposition*
- (-) Validation manly is based on peoples opinion *(Empirical)*
- (-) Coupling Criterions could be slithly different today *(Potentially outdated now)* including more or different criterions
  - Assume that they not change so fast, but this should be reevaluated